

SUMMARY

- First, I understood the project requirements and identified that the main objective of this project is **student data analysis, data modeling, and visualization using Power BI**.
- I collected and imported the given datasets into **Power BI Desktop**, which include:
 - Students data
 - Scores data
 - Attendance data
 - Behavior data
- After loading the data, I performed **basic data cleaning in Power Query**, such as:
 - Verifying and correcting data types
 - Removing unnecessary columns
 - Renaming columns for better clarity
 - After data cleaning, I moved to the **Model View** to organize and structure the tables properly.
 - The data model is designed using **fact tables and dimension tables**, which helps in creating an efficient and structured model.
- **Fact Tables:**
 - Scores Fact Table (exam scores and performance data)
 - Attendance Fact Table (present and absent records)
 - Behavior Fact Table (student behavior records)
- **Dimension Table:**
 - Students Dimension Table (student details like name, class, gender, and section)
 - I created **one-to-many relationships** from the Students dimension table to all fact tables using **StudentID** as the primary key.

- All relationships were kept **single-directional** to maintain a clean and logical model.
- After completing the data model, I created required **DAX measures** to calculate key metrics such as:
 - Total Students
 - Average Score and Average % Score
 - Attendance Percentage
 - Behavior Count by Type
 - Performance Category (High / Medium / Low)
 - I used **DAX functions** such as SUM, AVERAGE, CALCULATE, COUNT, DIVIDE, and SWITCH wherever required.
- An **interactive report** was then designed using:
 - KPI cards
 - Bar charts, line charts, and donut charts
 - Tables with conditional formatting
 - **Slicers** were added for Class, Section, Subject, and Term to allow dynamic filtering of data.
 - Advanced Power BI features such as **Drillthrough pages and custom Tooltips** were implemented to provide detailed and contextual insights.
 - Finally, the dashboard presents a **clear overview of academic performance, attendance patterns, and student behavior**, supporting data-driven academic analysis.
 - This project demonstrates my understanding of **Power BI data modeling, DAX calculations, data visualization, and report interactivity**.